

Challenges affecting the successful adoption of DevOps practices: A systematic literature review

Akalanka Nayanajith
Department of Industrial Management
University of Kelaniya
Sri Lanka.
98akalanka@gmail.com

Ruwan Wickramarachchi
Department of Industrial Management
University of Kelaniya
Sri Lanka.
ruwan@kln.ac.lk

Abstract—Contemporary business organizations are increasingly turning to IT-related tools, particularly software, to enhance the efficiency of their operational processes. As a result, time to deliver to market and the quality of software product has been a major concern to software development companies. Hence these organizations are willing to implement DevOps practices. DevOps has emerged as a promising approach in software development and operations, aiming to enhance collaboration, automation, and efficiency in delivering high-quality software products. While DevOps practices have gained significant attention globally, the successful adoption of DevOps practices in specific regions, may encounter unique challenges for various reasons. It also has become a critical endeavor for organizations in the developing countries, like Sri Lanka, seeking to enhance their software development and delivery processes. This systematic literature review explores the challenges affecting the successful adoption of DevOps practices by extensively analyzing relevant scholarly articles, conference papers, and industry reports. The identified challenges encompass organizational culture, technical infrastructure, skill gaps, regulatory compliance, and resource constraints. More specifically, the analysis revealed major challenges such as resistance to change, absence of supportive and visionary leadership, outdated legacy systems, high cost of investment, skills gaps, communication gaps between development and operations teams, and a deep-seated company culture.

Keywords – DevOps, DevOps adoption, DevOps challenges

I. INTRODUCTION

The global IT industry has witnessed exponential growth in recent years driven by the rapid pace of technological innovation and digital transformation across business sectors. Global IT spending is projected to reach \$4.5 trillion in 2024, a substantial increase of 5.1% from 2023, with software emerging as the fastest growing segment expected to grow 10.5% year on year [1]. This surge can be attributed to the rising need for cutting-edge software solutions that can address the evolving challenges and opportunities in the digital landscape.

However, traditional software development approaches, such as the waterfall methodology with its sequential process, have proven inadequate for the rapid iteration and continuous delivery required in today's fast-paced market [2]. These methodologies often lead to long development cycles, limited flexibility, and difficulties in adapting to changing requirements.

To overcome these limitations, Agile methodology was widely adopted by the software industry. Embracing adaptive planning, continuous improvement and rapid iterations, Agile enabled faster time-to-market and enhanced collaboration with stakeholders [3]. However, lack of cooperation between development and operations teams posed challenges. DevOps methodology emerged to bridge this gap by fostering

integration, collaboration and automation across the software development lifecycle [4].

DevOps, a portmanteau of “development” and “operations”, facilitates continuous development cycles enabling faster, reliable and efficient software delivery leveraging practices like continuous integration (CI), continuous delivery (CD) and continuous deployment. In CI, code changes are frequently integrated and build automation ensures rapid feedback on integration errors [5]. CD automates the software release process allowing rapid, frequent deployments while ensuring quality through test automation.

The adoption of DevOps practices has yielded significant benefits for global organizations, with industry leaders such as Amazon, Netflix and Google have extensively adopted DevOps, attributing significant improvements in productivity, stability and faster innovation to this approach [6].

Sri Lanka's position as an emerging IT/BPM destination has driven significant interest in DevOps adoption. Software export revenue grew by 11% in 2022 to an estimated USD 1.6 billion based on SLASSCOM estimates.

However, challenges persist in the adoption of DevOps including lack of technical expertise in DevOps practices, complexity of legacy modernization, organizational culture gaps and communication barriers. Studies have indicated a noticeable resistance to change along with high costs as key barriers to DevOps adoption in Sri Lanka [7],[8]. This paper aims to systematically analyze these challenges affecting the successful adoption of DevOps practices through comprehensive literature analysis providing actionable insights to key stakeholders in the Sri Lankan software industry.

II. METHODOLOGY

The PRISMA approach was utilized in this systematic literature review to ensure a rigorous and reliable process [9]. Relevant academic databases such as IEEE Xplore, Google Scholar, and ResearchGate were searched to retrieve literature related to DevOps adoption challenges. The search used targeted keywords including "DevOps," "DevOps adoption," and "DevOps challenges." Appropriate search operators like "AND" and "OR" were employed to connect the keywords and refine the search.

The initial search retrieved 21,327 documents. After screening the titles for relevance, 67 documents were selected for further review. The full texts of these documents were obtained and thoroughly analyzed based on predefined inclusion and exclusion criteria. Specifically, only English language research publications from 2010-2023 were considered (16 non-English articles were excluded).

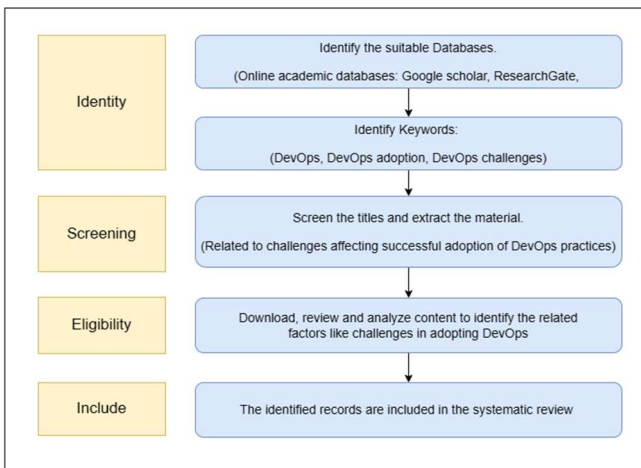


Fig.1. Methodology of the literature review

The types of publications were limited to peer-reviewed research articles, conference papers, and journal publications. Book chapters, unpublished manuscripts, and non-academic sources were excluded (Book chapters-3, unpublished manuscripts-8, non-academic sources-19. Total of 30 articles has been excluded).

During the full-text review, the focus was on identifying papers that investigated the challenges affecting successful DevOps adoption. Ultimately 21 research articles that directly examined impediments and critical success factors related to implementing DevOps practices in Sri Lanka were selected.

The selected articles provided the basis for a systematic synthesis of the current literature around DevOps adoption challenges in the Sri Lanka setting. By adhering to the rigorous PRISMA methodology, this literature review achieved a standardized, unbiased, and transparent process. The insights derived from this review will illuminate the obstacles experienced by Sri Lankan organizations in transitioning to DevOps.

III. RESULTS AND ANALYSIS

Analysis has revealed several aspects of the DevOps implementation and the challenges that affects the adoption.

A. DevOps adoption in organizations

As technology continues to advance at a rapid pace, companies across a wide range of industries, especially those in the information technology sector, feel increasing pressure to adapt their business strategies and operations in order to achieve higher productivity and deliver high-quality products and services to their customers [10]. This is particularly true for software companies, who must frequently update and release new versions of their software applications in order to meet the ever-evolving needs and expectations of users, who are demanding applications that are more user-friendly, intuitive, and easier to use. [7].

However, the need for frequent software updates and new releases also introduces major quality assurance challenges. Each deployment risks introducing new bugs and defects that could frustrate customers and require additional time and resources to fix. Without strong quality assurance practices in place, this fast pace of continuous delivery could lead to

serious customer dissatisfaction, wasted expenditures, and damage to a company's reputation.

Implementing DevOps practices can help software teams overcome these challenges by promoting four key principles: source control, test automation, cross-functional teams, and continuous delivery [11]. Source control through systems like Git allows multiple contributors to collaborate on software projects transparently, track changes, catch issues early, and revert back to previous versions if needed [12]. Comprehensive test automation ensures that new features and bug fixes are thoroughly validated across a wide range of devices and scenarios before reaching end users. Building cohesive teams where developers, quality engineers, security experts, and operations staff share ownership of the entire delivery lifecycle rather than operating in siloes promotes accountability. And automating continuous delivery through DevOps tools like Jenkins eliminates manual errors, speeds up release cycles, and provides faster user feedback [13].

By adopting DevOps culture, processes, and technologies, software organizations can achieve the pace and quality of innovation needed to satisfy customers in an increasingly competitive, fast-moving market. The practices not only accelerate release velocity but also provide the configuration control, visibility, collaboration and quality assurance required when deploying updates on a continuous basis [11].

B. Challenges in DevOps adoption

There are many challenges related to DevOps identified through the literature. The scarcity of skilled and technically proficient staff poses a primary challenge, as organizations struggle to recruit and retain qualified DevOps professionals. Additionally, the steep learning curve associated with new tools and processes hampers progress, necessitating robust training programs [14], [15]. Limited investments in skill development and training by most companies further exacerbates talent shortages.

Resistance to change and uncertainty constitute recurring adoption barriers, as employees habitually resisting alterations introduced by DevOps transitions [15], [16]. Shifting responsibilities and volumes of change contribute to resistance and uncertainty, underscoring the need for cultural mindset shifts [17]. Resistance often stems from sentiments surrounding job security, comfort with the status quo, and change fatigue [18].

Infrastructure challenges [24] highlight the essential role of a well-designed common infrastructure, with issues arising in design, hardware and software problem-solving, and the provisioning necessary for effective DevOps execution. An organization's maturity and size frequently impact DevOps integration. Larger corporations with complex legacy systems hamper tooling and test automation [11], [19]. Transitioning from entrenched and outdated platforms proves cumbersome, demanding significant upfront resource investment [13], [20]. Siloed communication channels also delay collaboration between teams playing radically different roles than before [6], [21].

The challenges identified in the research conducted by Battina et al. in 2020 offer valuable insights on key obstacles in adopting DevOps practices during software development. This paper emphasizes the difficulty software development teams face in balancing client expectations, functionality, and

software quality, especially with the pressure for fast delivery and short development cycles inherent in DevOps. The challenge of maintaining robust security measures becomes apparent, as accelerated timelines may compromise thorough security evaluations, leading to potential security risks. Communication issues between development and security teams are also identified, causing confusion, delayed deliveries, and frustration [12].

Visionary leadership and enhanced cross-functional coordination are imperative to mitigate these multifaceted challenges. Strong leadership plays a pivotal role in addressing change resistance, realigning company culture, and promoting collaboration between siloed departments [25]. Lack of DevOps understanding amongst executives and pervasive legacy mindsets contribute to adoption barriers [20]. Breaking down communication silos is equally vital for successful adoption [12].

The research studies conducted by Riungu-Kalliosaari et al. in 2016 [16] and Senapathi et al. in 2018 [15] offer valuable insights into the challenges associated with DevOps adoption. They identify challenges stemming from communication patterns, organizational culture, technical constraints, and the ambiguous nature of the DevOps concept. The insufficient communication between development and operations teams emerges as a critical obstacle, hindering the seamless integration of DevOps practices. Cultural resistance to change further complicates the adoption process, as DevOps necessitates profound shifts in organizational mindset and culture. Technical challenges related to tool integration and automation add complexity to the implementation journey. Additionally, the lack of a clear and shared understanding of the meaning of DevOps can lead to confusion and hinder the adoption process.

Maroukian and Gulliver's exploration [21] sheds light on the link between leadership and DevOps practice and principle adoption. Challenges identified include poor communication and information flow, deep-seated company culture, and operations not participating in requirements specifications. The study emphasizes the role of leadership in overcoming these challenges, suggesting the need for cross-functional leadership roles, influential skills, holistic systems thinking, and increased awareness around dealing with suboptimal productivity.

While these research studies elaborate on various DevOps adoption challenges, as a summary the following obstacles are consistently mentioned across all of them.

- Resistance to change
- Legacy systems
- Cost of investment
- Lack of skills
- Communication gaps between development and operations teams
- Deep-seated company culture

In a related vein, Zarour et al. in 2020 [12] identify a range of challenges in the adoption of the DevOps process model in Saudi Arabia. Cultural resistance, lack of skills, legacy systems, security and compliance concerns, toolchain complexity, and resistance to change are outlined as significant hurdles. Trigo's insights from a large European Telco [19] echo similar sentiments, emphasizing resistance to change, organizational size, maturity, cost of investment, and

the lack of technical skills as common challenges faced by organizations embracing DevOps. The constant evolution of processes inherent to the DevOps culture is also identified as a challenge, requiring organizations to remain adaptable and ready to adjust whenever necessary [19]. These studies indicate that software development organizations face a common set of challenges as well as region-specific ones. Together, these studies contribute valuable insights into the nuanced challenges encompassing human behavior, organizational culture, and leadership dynamics in the context of DevOps adoption.

There are numerous challenges hindering the widespread adoption of DevOps practices within the Sri Lankan IT industry. The literature on DevOps challenges provides a comprehensive understanding of the multifaceted obstacles organizations encounter in their adoption journey. Arsecularatne and Wickramarachchi's study in 2023 [20] and Jayakody et al in 2022 [22] delve into human-centric behavioral challenges, underscoring issues such as the lack of a DevOps mindset, resistance to change, communication and collaboration gaps between teams, and the absence of authority for DevOps engineers. This human-centric perspective aligns with the broader challenges outlined by Arsecularatne and Wickramarachchi, who categorize DevOps challenges into four main domains: Automation, Measurement, Sharing, and Culture [20].

In addition to these behavioral challenges, legacy tools and applications pose significant hurdles, creating conflicts with platforms and libraries, introducing difficulties in handling automation and security, and necessitating a time-consuming process for creating automation that demands a substantial upfront investment of time and resources. The study accentuates the profound impact of these challenges on software delivery, emphasizing the intricate interplay between human behavior, legacy systems, and organizational culture [20], [22].

By categorizing challenges, Arsecularatne and Wickramarachchi's framework offers a structured approach to understanding the multifaceted nature of obstacles faced during DevOps adoption. The challenges identified, such as overcoming separate mindsets of Dev versus Ops, transitioning from legacy infrastructure to microservices, and addressing reluctance to change, provide a roadmap for organizations in the Sri Lankan IT industry to systematically address and navigate the complexities associated with DevOps implementation.

C. Mitigation Strategies

To address the above-mentioned challenges, Battina underscores the importance of integrating security measures early in the software development lifecycle, promoting collaboration, and employing automated tools for rapid security checks. Automation is seen as a critical element in mitigating risks associated with human errors and vulnerabilities [13]. The Maroukian and Gulliver's research underscores the critical role of leadership in the successful adoption of DevOps practices and principles, emphasizing the necessity for organizations to invest in developing the leadership skills and capabilities required to overcome adoption barriers [21].

Effective leadership should make the cultural transition and team collaboration a priority to drive DevOps adoption [17]. Providing high-quality training in automation,

continuous practices, and conducting knowledge-sharing workshops can help upskill teams [15]. Starting small, with pilot DevOps implementations, and getting quick wins can demonstrate value and reduce uncertainty around changes [12]. Arsecularatne and Wickramarachchi's research suggests that adopting chaos engineering is much effective way to overcome the challenges [20]. Careful legacy system integration planning and modular architecture approaches can reduce complexities [5]. With disciplined mitigation strategies, the impact of DevOps adoption challenges can be minimized.

IV. DISCUSSION

The adoption of DevOps practices in software development companies face some key challenges that can impact the ability of organizations to gain the intended benefits. One major challenge identified is the lack of necessary skills and expertise among professionals to effectively implement critical DevOps tasks. This shortage of technical competency hinders progress and slows down DevOps adoption. Relatedly, the complexity of legacy systems and frameworks also poses difficulties in integration with modern DevOps tools and processes

Additionally, cultural challenges like resistance to change and lack of collaboration between teams are recurring barriers. The changes to tools, processes, and responsibilities brought on by DevOps can create uncertainty that leads to resistance among employees. Overcoming these deeply ingrained cultural mindsets requires effective leadership and communication across the organization.

Considering Sri Lankan context, only 2 relevant researches has been found, which crucially highlights the need for empirical research to further validate and explore these challenges within the country. As DevOps adoption continues to gain momentum, and Sri Lanka is becoming major software developing hub, investigating the challenges which affects the adoption of DevOps could provide insights into mitigating some of these implementation challenges. The challenges found in previous literature can be different as there is a huge brain drain in recent past and because of the economic crisis happened in 2022.

Apart from the lack of research papers, no quantitative research has been done in Sri Lanka. It is important to validate the challenges through hypothesis testing as validating the challenges quantitatively rather than qualitatively will provide more objective results, given that qualitative findings can be subject to bias. Furthermore, hypothesis testing will enable statistical analysis to identify the factors that impact results the most within the Sri Lankan context. The current shortage of quantitative studies presents an opportunity for impactful research in this area.

V. CONCLUSION

The adoption of DevOps within the rapidly evolving global IT sector faces obstacles like a shortage of skills, cultural resistance, and integration complexities. However, recognition of DevOps' benefits has driven efforts to implement its practices more widely. The availability of professionals with strong technical competencies along with leadership commitment to transform embedded mindsets and processes are key to overcoming these barriers. Further empirical research can shed light on these challenges globally

and explore approaches to enable successful DevOps adoption across regions. In particular, there is a shortage of localized research investigating these issues in specific countries. Conducting quantitative studies locally, like in Sri Lanka, is important to measure the magnitude of each obstacle and understand nuances in that context. While global challenges exist, focused research within countries can complement broader findings and provide actionable insights tailored to the needs of local IT sectors. More localized empirical research will allow comparison and identification of specialized strategies to accelerate DevOps adoption in specific national contexts.

REFERENCES

- [1] IDC (2023). Worldwide ICT Spending Guide Industry and Company Size. Retrieved from <https://www.idc.com/getdoc.jsp?containerId=prUS49512122>
- [2] Smeds, J., Nybom, K., & Porres, I. (2015). DevOps: a definition and perceived adoption impediments. In *International Conference on Agile Software Development* (pp. 166-177). Springer, Cham.
- [3] Schwalbe, K. (2016). *Information technology project management* (Eighth edition.). Boston, MA: Course Technology, Cengage Learning.
- [4] Shahin, M., Ali Babar, M., & Zhu, L. (2017). Continuous integration, delivery and deployment: A systematic review on approaches, tools, challenges and practices. *IEEE Access*, 5, 3909-3943.
- [5] Humble, J., & Farley, D. (2010). *Continuous delivery: reliable software releases through build, test, and deployment automation*. Addison-Wesley Professional.
- [6] Khan, M. S., Khan, A. W., Khan, F., Khan, M. A., & Whangbo, T. K. (2022). Critical challenges to adopt DevOps culture in software organizations: A systematic review. *IEEE Access*, 10, 14339-14349.
- [7] Jones, S., Noppen, J., & Lettice, F. (2016, July). Management challenges for DevOps adoption within UK SMEs. In *Proceedings of the 2nd International Workshop on quality-aware devops* (pp. 7-11).
- [8] Jayasinghe, S., & Wijayarathna, D. (2020). DevOps implementation challenges in Sri Lankan software development industry. *International Journal of Advanced Research in Computer Science and Software Engineering*, 10(5), 94-100.
- [9] A. Liberati et al., "The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration," *BMJ*, vol. 339, no. jul21 1, p. b2700, 2009.
- [10] Toh, M. Z., Sahibuddin, S., & Mahrin, M. N. R. (2019, February). Adoption issues in DevOps from the perspective of continuous delivery pipeline. In *Proceedings of the 2019 8th International Conference on Software and Computer Applications* (pp. 173-177).
- [11] Leite, L., Rocha, C., Kon, F., Milojicic, D., & Meirelles, P. (2019). A survey of DevOps concepts and challenges. *ACM Computing Surveys (CSUR)*, 52(6), 1-35.
- [12] Zarour, M., Alhammad, N., Alenezi, M., & Alsarayrah, K. (2020). DevOps process model adoption in Saudi Arabia: an empirical study. *Jordanian Journal of Computers and Information Technology*, 6(3).
- [13] Battina, D. S. (2021). The Challenges and Mitigation Strategies of Using DevOps during Software Development. *International Journal of Creative Research Thoughts (IJCRT)*, ISSN, 2320-2882.
- [14] L. Chen, "Continuous Delivery at Scale: Challenges and Opportunities," 2018 IEEE/ACM 4th International Workshop on Rapid Continuous Software Engineering (RCoSE), Gothenburg, Sweden, 2018, pp. 42-42.
- [15] Senapathi, M., Buchan, J., & Osman, H. (2018). DevOps Capabilities, Practices, and Challenges: Insights from a Case Study. *Proceedings of the 22nd International Conference on Evaluation and Assessment in Software Engineering 2018*.
- [16] Riungu-Kalliosaari, L., Mäkinen, S., Lwakatere, L. E., Tiihonen, J., & Männistö, T. (2016). DevOps adoption benefits and challenges in practice: A case study. In *Product-Focused Software Process Improvement: 17th International Conference, PROFES 2016, Trondheim, Norway, November 22-24, 2016, Proceedings 17* (pp. 590-597). Springer International Publishing.

- [17] Díaz, Jessica & López-Fernández, Daniel & Pérez-Martínez, Jorge & González-Prieto, Ángel. (2021). Why are many businesses instilling a DevOps culture into their organization?. *Empirical Software Engineering*. 26. 10.1007/s10664-020-09919-3.
- [18] Krey, M. (2022). Devops adoption: challenges & barriers.
- [19] Trigo, A., Varajão, J., & Sousa, L. (2022). DevOps adoption: Insights from a large European Telco. *Cogent Engineering*, 9(1), 2083474.
- [20] Arsecularatne, M., & Wickramarachchi, R. (2023, June). Adoptability of Chaos Engineering with DevOps to Stimulate the Software Delivery Performance. In *2023 International Research Conference on Smart Computing and Systems Engineering (SCSE)* (Vol. 6, pp. 1-8). IEEE.
- [21] Maroukian, K., & Gulliver, S. (2020). Exploring the link between leadership and Devops practice and principle adoption. *Advanced Computing: An International Journal*, 11(4).
- [22] Jayakody, J.A.V.M.K & Wijayanayake, Janaka. (2021). Challenges for adopting DevOps in information technology projects. 203-210. 10.1109/SCSE53661.2021.9568348.
- [23] Shameem, M. (2022). A systematic literature review of challenges factors for implementing DevOps practices in software development organizations: A development and operation teams perspective. *Evolving Software Processes: Trends and Future Directions*, 187-199.
- [24] Azad, N., & Hyrynsalmi, S. (2023). DevOps critical succes factors—A systematic literature review. *Information and Software Technology*, 107150.
- [25] Azad, N. (2022, May). Understanding DevOps critical success factors and organizational practices. In *Proceedings of the 5th International Workshop on Software-intensive Business: Towards Sustainable Software Business* (pp. 83-90).
- [26] R. K. Gupta, M. Venkatachalapathy and F. K. Jeberla, "Challenges in Adopting Continuous Delivery and DevOps in a Globally Distributed Product Team: A Case Study of a Healthcare Organization," 2019 ACM/IEEE 14th International Conference on Global Software Engineering (ICGSE), Montreal, QC, Canada, 2019, pp. 30-34, doi: 10.1109/ICGSE.2019.00020.